14.2.9 Enzymes:

There are many biochemical reactions that occur in our living cells. Digestion of food and harvesting the energy from them, and synthesis of necessary molecules required for various cellular functions are examples for such reactions. All these reactions are catalysed by special proteins called enzymes. These biocatalysts accelerate the reaction rate in the orders of 10^5 and also make them highly specific. The high specificity is followed allowing many reactions to occur within the cell. For example, the *Carbonic anhydrase* enzyme catalyses the interconversion of carbonic acid to water and carbon dioxide. Sucrase enzyme catalyses the hydrolysis of sucrose to fructose and glucose. Lactase enzyme hydrolyses the lactose into its constituent monosaccharides, glucose and galactose.

14.2.10 Mechanism of enzyme action:

Enzymes are biocatalysts that catalyse a specific biochemical reaction. They generally activate the reaction by reducing the activation energy by stabilising the transition state. In a typical reaction enzyme (E) binds with the substrate (S) molecule reversibly to produce an enzyme-substrate complex (ES). During this stage the substrate is converted into product and the enzyme becomes free, and ready to bind to another substrate molecule. More detailed mechanism is discussed in the unit XI surface chemistry.

$$\underset{Enzyme}{E} + \underset{Substate}{S} \rightleftharpoons \underset{Enzyme\text{ - substrate complex}}{[ES]}$$

$$[ES] \rightarrow E+P$$

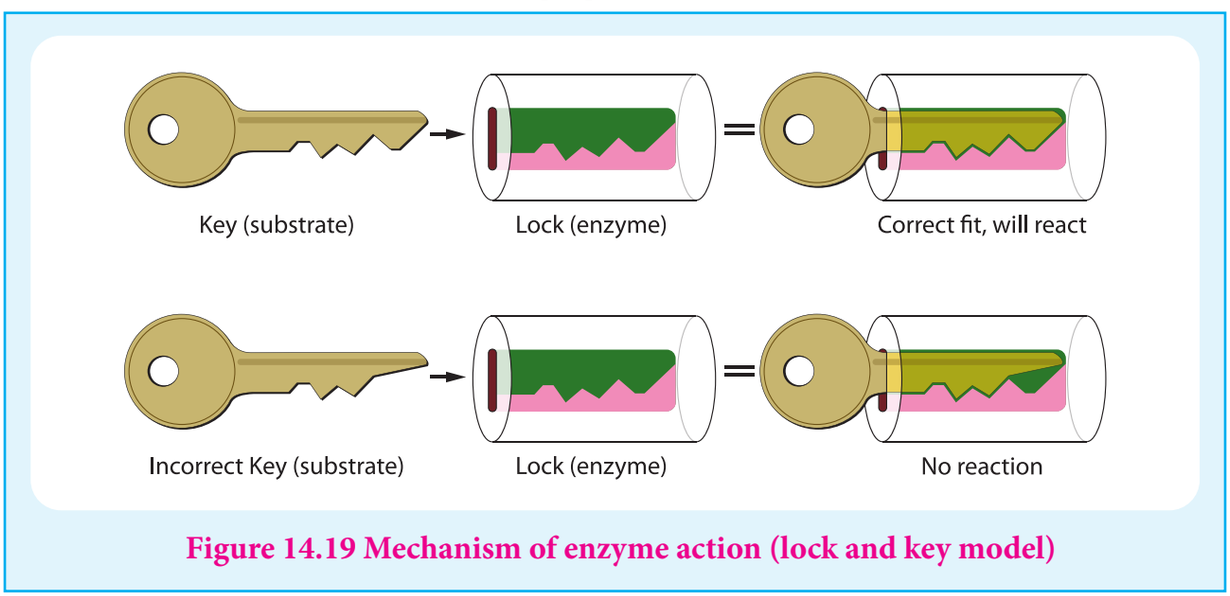


Figure 14.19 Mechanism of enzyme action (lock and key model)

14.3 Lipids:

Lipids are organic molecules that are soluble in organic solvents such as chloroform and methanol and insoluble in water. The word lipid is derived from the Greek word *'lipos'* meaning fat. They are the principal components of cell membranes. In addition, they also act as energy source for living systems. Fat provides 2-3 fold higher energy compared to carbohydrates / proteins.

XII U14-Biomolecules_New.indd 257 11-01-2021 16:38:03